

COMP 1324 Algorithmics

Tutorial 10

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- 1 Consider the optimisation problem below. Find the solution using the simplex algorithm and graphically. Does the problem have more than one solution?

$$\begin{aligned} &\text{maximize} && 6x_1 + 3x_2 \\ &\text{subject to} && 2x_1 + x_2 \leq 20 \\ &&& x_1 - x_2 \leq 4 \\ &&& 3x_1 + 2x_2 \leq 35 \\ &&& x_1, x_2 \geq 0 \end{aligned}$$

Linear programming

	x_1	x_2	x_3	x_4	x_5	RHS
z	-6	-3	0	0	0	0
x_3	2	1	1	0	0	20
x_4	1	-1	0	1	0	4
x_5	3	2	0	0	1	35

Pivot: x_1 entering and x_4 leaving.

	x_1	x_2	x_3	x_4	x_5	RHS
z	0	-9	0	6	0	24
x_3	0	3	1	-2	0	12
x_1	1	-1	0	1	0	4
x_5	0	5	0	-3	1	23

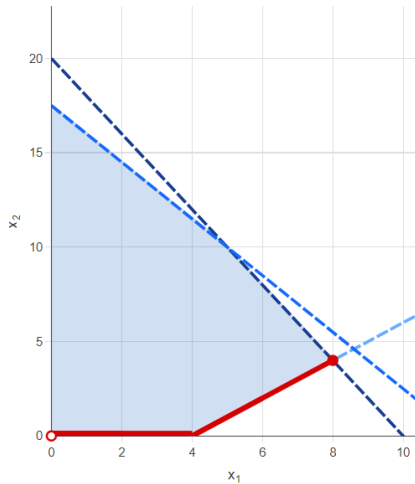
	x_1	x_2	x_3	x_4	x_5	RHS
z	0	-9	0	6	0	24
x_3	0	3	1	-2	0	12
x_1	1	-1	0	1	0	4
x_5	0	5	0	-3	1	23

Pivot: x_2 entering and x_3 leaving.

	x_1	x_2	x_3	x_4	x_5	RHS
z	0	0	3	0	0	60
x_2	0	1	$1/3$	$-2/3$	0	4
x_1	1	0	$1/3$	$1/3$	0	8
x_5	0	0	$-5/3$	$1/3$	1	3

Since all coefficients of row 0 are positive simplex stops. From the table we get: $z = 60, x_1 = 8, x_2 = 4$ Graphically

Graphically



Note that the constraint lines: $2x_1 + x_2 = 20$ and $3x_1 + 2x_2$ intersect at $x_1 = 5$ and $x_2 = 10$. At that point $(5, 10)$ the objective function has the value 60 also. It means the consecutive vertices $(8, 4)$ and $(5, 10)$ have the same maximum value for the objective function. By the fundamental theorem of LP we know all points on that segment are also solutions.

An easier way to detect multiple solutions is that if one of the constraints is "parallel" to the objective function. In this case $6x_1 + 3x_2$ is parallel to $2x_1 + x_2$.

Question 2

- 1 Solve the following LP using the simplex algorithm.

$$\begin{aligned} &\text{maximize} && 5x_1 + 3x_2 \\ &\text{subject to} && 2x_1 + x_2 \leq 20 \\ &&& x_1 - x_2 \leq -16 \\ &&& 3x_1 + 2x_2 \leq 35 \\ &&& x_1, x_2 \geq 0 \end{aligned}$$

Question 2 solution

Since one of the constants is negative we need to check if the problem is feasible and if so obtain an initial feasible solution. We start the 2 phase simplex.

$$\begin{aligned} &\text{maximize} && -x_0 \\ &\text{subject to} && 2x_1 + x_2 - x_0 + x_3 = 20 \\ & && x_1 - x_2 - x_0 + x_4 = -16 \\ & && 3x_1 + 2x_2 - x_0 + x_5 = 35 \\ & && x_i \geq 0, i = 0 \dots 5 \end{aligned}$$

Question 2 solution

First step is to pivot with x_0 as entering variable and x_4 (row with "most" negative b) as leaving variable

	x_1	x_2	x_0	x_3	x_4	x_5	RHS
z	0	0	1	0	0	0	0
x_3	2	1	-1	1	0	0	20
x_4	1	-1	-1	0	1	0	-16
x_5	3	2	-1	0	0	1	35

pivot $x_0 \leftrightarrow x_4$

	x_1	x_2	x_0	x_3	x_4	x_5	RHS
z	1	-1	0	0	1	0	-16
x_3	1	2	0	1	-1	0	36
x_0	-1	1	1	0	-1	0	16
x_5	2	3	0	0	-1	1	51

Next pivot $x_2 \leftrightarrow x_0$

Question 2 solution

	x_1	x_2	x_0	x_3	x_4	x_5	RHS
z	0	0	1	0	0	0	0
x_3	3	0	-2	1	1	0	4
x_2	-1	1	1	0	-1	0	16
x_5	5	0	-3	0	2	1	3

- All coefficients of z are positive so simplex terminates. $z = 0$ so solution is feasible.
- We use the above as a starting solution for phase 2 while dropping x_0 and using the original objective function
- Also, the objective function should contain non-basic variables only. The initial objective function was $5x_1 + 3x_2$ and x_2 is a basic variable so we replace it $x_2 = 16 + x_1 + x_4$. Finally, the new objective function is $5x_1 + 3(16 + x_1 + x_4) = 8x_1 + 48 + 3x_4$.

Question 2 solution

Phase 2 starts from the following tableau

	x_1	x_2	x_3	x_4	x_5	RHS
z	-8	0	0	-3	0	48
x_3	3	0	1	1	0	4
x_2	-1	1	0	-1	0	16
x_5	5	0	0	2	1	3

Pivot: $x_1 \leftrightarrow x_5$

	x_1	x_2	x_3	x_4	x_5	RHS
z	0	0	0	$1/5$	$8/5$	$264/5$
x_3	0	0	1	$-1/5$	$-3/5$	$11/5$
x_2	0	1	0	$-3/5$	$1/5$	$83/5$
x_1	1	0	0	$2/5$	$1/5$	$3/5$

All coefficients of z are positive so simplex terminates. The solution is $z = 264/5 = 52.8$, $x_1 = 3/5 = 0.6$, $x_2 = 83/5 = 16.6$.

Question 3

Convert into standard form

$$\begin{array}{ll}\text{minimise} & 2x_1 + 7x_2 + x_3 \\ \text{subject to} & x_1 - x_3 = 7 \\ & 3x_1 + x_2 \geq 24 \\ & x_2 \geq 0 \\ & x_3 \leq 0\end{array}$$

Question 3 solution

Convert min to max

$$\begin{aligned} &\text{maximise} && -2x_1 - 7x_2 - x_3 \\ &\text{subject to} && x_1 - x_3 = 7 \\ &&& 3x_1 + x_2 \geq 24 \\ &&& x_2 \geq 0 \\ &&& x_3 \leq 0 \end{aligned}$$

Question 3 solution

replace x_1 by $x_1^+ - x_1^-$

$$\begin{aligned} &\text{maximise} && -2x_1^+ + 2x_1^- - 7x_2 - x_3 \\ &\text{subject to} && x_1^+ - x_1^- - x_3 = 7 \\ &&& 3x_1^+ - 3x_1^- + x_2 \geq 24 \\ &&& x_1^+ \geq 0 \\ &&& x_1^- \geq 0 \\ &&& x_2 \geq 0 \\ &&& x_3 \leq 0 \end{aligned}$$

Question 3 solution

replace = by $\geq$ and $\leq$

$$\begin{aligned} &\text{maximise} && -2x_1^+ + 2x_1^- - 7x_2 - x_3 \\ &\text{subject to} && x_1^+ - x_1^- - x_3 \leq 7 \\ &&& x_1^+ - x_1^- - x_3 \geq 7 \\ &&& 3x_1^+ - 3x_1^- + x_2 \geq 24 \\ &&& x_1^+ \geq 0 \\ &&& x_1^- \geq 0 \\ &&& x_2 \geq 0 \\ &&& x_3 \leq 0 \end{aligned}$$

Question 3 solution

replace $\geq$ by $\leq$

$$\begin{aligned} &\text{maximise} && -2x_1^+ + 2x_1^- - 7x_2 - x_3 \\ &\text{subject to} && x_1^+ - x_1^- - x_3 \leq 7 \\ &&& -x_1^+ + x_1^- + x_3 \leq -7 \\ &&& -3x_1^+ + 3x_1^- - x_2 \leq -24 \\ &&& x_1^+, x_1^-, x_2 \geq 0 \\ &&& x_3 \leq 0 \end{aligned}$$

Question 3 solution

replace x_3 by $-x_3$

$$\begin{aligned} &\text{maximise} && -2x_1^+ + 2x_1^- - 7x_2 + x_3 \\ &\text{subject to} && x_1^+ - x_1^- + x_3 \leq 7 \\ &&& -x_1^+ + x_1^- - x_3 \leq -7 \\ &&& -3x_1^+ + 3x_1^- - x_2 \leq -24 \\ &&& x_1^+, x_1^-, x_2, x_3 \geq 0 \end{aligned}$$

Question 3 solution

replace $x_1^+ \leftrightarrow x_1$ and $x_1^- \leftrightarrow x_4$

$$\begin{aligned} &\text{maximise} && -2x_1 + 2x_4 - 7x_2 + x_3 \\ &\text{subject to} && x_1 - x_4 + x_3 \leq 7 \\ &&& -x_1 + x_4 - x_3 \leq -7 \\ &&& -3x_1 + 3x_4 - x_2 \leq -24 \\ &&& x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Question 4

Show that the following is infeasible

maximise $3x_1 - 2x_2$

subject to

$$x_1 + x_2 \leq 2$$

$$-2x_1 - 2x_2 \leq -10$$

$$x_1, x_2 \geq 0$$

Question 4 solution

Add variable x_0 and use phase 1

$$\begin{aligned} &\text{maximise} && -x_0 \\ &\text{subject to} && x_1 + x_2 - x_0 \leq 2 \\ &&& -2x_1 - 2x_2 - x_0 \leq -10 \\ &&& x_0, x_1, x_2 \geq 0 \end{aligned}$$

Convert it to slack form

$$\begin{aligned} &\text{maximise} && -x_0 \\ &\text{subject to} && x_1 + x_2 - x_0 + x_3 = 2 \\ &&& -2x_1 - 2x_2 - x_0 + x_4 = -10 \\ &&& x_0, x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Question 4 solution

In tableau form

	x_1	x_2	x_0	x_3	x_4	RHS
z	0	0	1	0	0	0
x_3	1	1	-1	1	0	2
x_4	-2	-2	-1	0	1	-10

In the auxiliary problem the first pivot is x_0 entering and the "most" negative RHS (x_4) leaving. After the first pivot we get

	x_1	x_2	x_0	x_3	x_4	RHS
z	-2	-2	0	0	1	-10
x_3	3	3	0	1	-1	12
x_0	2	2	1	0	-1	10

Question 4 solution

x_1 entering and x_3 leaving

	x_1	x_2	x_0	x_3	x_4	RHS
z	-2	-2	0	0	1	-10
x_3	3	3	0	1	-1	12
x_0	2	2	1	0	-1	10

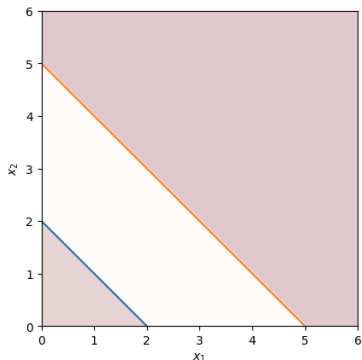
After performing the pivot we get

	x_1	x_2	x_0	x_3	x_4	RHS
z	0	0	0	$2/3$	$1/3$	-2
x_1	1	1	0	$1/3$	$-1/3$	4
x_0	0	0	1	$-2/3$	$-1/3$	2

All the coefficients of z are positive so simplex terminates. The RHS of z is non-zero so the solution is infeasible.

Question 4 solution

We can see the infeasibility of the problem graphically.



Question 5

Consider the following LP

$$\begin{array}{ll}\text{maximise} & -x_1 - x_2 \\ \text{subject to} & -2x_1 - x_2 \leq 4 \\ & -2x_1 + 4x_2 \leq -8 \\ & -x_1 + 3x_2 \leq -7 \\ & x_1, x_2 \geq 0\end{array}$$

What is the dual of the above LP problem?

Question 5 (a)

$$\begin{array}{ll}\text{minimise} & 4y_1 - 8y_2 - 7y_3 \\ \text{subject to} & -2y_1 - 2y_2 - y_3 \geq -1 \\ & -y_1 + 4y_2 + 3y_3 \geq -1 \\ & y_1, y_2, y_3 \geq 0\end{array}$$

convert it to standard form

$$\begin{array}{ll}\text{maximise} & -4y_1 + 8y_2 + 7y_3 \\ \text{subject to} & 2y_1 + 2y_2 + y_3 \leq 1 \\ & y_1 - 4y_2 - 3y_3 \leq 1 \\ & y_1, y_2, y_3 \geq 0\end{array}$$

Question 5 (b)

The primal in slack form

$$\begin{array}{ll}\text{maximise} & -x_1 - x_2 \\ \text{subject to} & -2x_1 - x_2 + x_3 = 4 \\ & -2x_1 + 4x_2 + x_4 = -8 \\ & -x_1 + 3x_2 + x_5 = -7 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0\end{array}$$

The dual in slack form

$$\begin{array}{ll}\text{minimise} & 4y_1 - 8y_2 - 7y_3 \\ \text{subject to} & -2y_1 - 2y_2 - y_3 + y_4 = -1 \\ & -y_1 + 4y_2 + 3y_3 + y_5 = -1 \\ & y_1, y_2, y_3, y_4, y_5 \geq 0\end{array}$$

Question 5 (c)

To solve the dual we turn it into a maximisation problem

$$\begin{aligned} &\text{maximise} && -4y_1 + 8y_2 + 7y_3 \\ &\text{subject to} && 2y_1 + 2y_2 + y_3 + y_4 = 1 \\ &&& y_1 - 4y_2 - 3y_3 + y_5 = 1 \\ &&& y_1, y_2, y_3, y_4, y_5 \geq 0 \end{aligned}$$

tableau form

	y_1	y_2	y_3	y_4	y_5	RHS
z	4	-8	-7	0	0	0
y_4	2	2	1	1	0	1
y_5	1	-4	-3	0	1	1

Pivot: $y_2 \leftrightarrow y_4$

Question 5 (c)

	y_1	y_2	y_3	y_4	y_5	RHS
z	12	0	-3	4	0	4
y_2	1	1	1/2	1/2	0	1/2
y_5	5	0	-1	2	1	3

Pivot: y_3 entering and y_2 leaving

	y_1	y_2	y_3	y_4	y_5	RHS
z	18	6	0	7	0	7
y_3	2	2	1	1	0	1
y_5	7	2	0	3	1	4

Therefore, for the dual, $\max(-z)$ is 7 so $\min(z)$ is -7 which by the strong duality theorem is the same as the primal.

We need the values of the variables for the next parts:

$$y_1 = 0, y_2 = 0, y_3 = 1, y_4 = 0, y_5 = 4.$$

Question 5 (d)

Complementary slackness. Let $x_1, \dots, x_n$ and $u_1, \dots, u_m$ be the variables/slack variables for the primal and $y_1, \dots, y_m$ and $w_1, \dots, w_n$ the variables/slack variables for the dual.

Complementary slackness states that if both the primal and dual are feasible then $x_i w_i = 0$ and $y_j u_j = 0$ for all $i = 1 \dots n, j = 1 \dots m$.

Recall that $y_1 = 0, y_2 = 0, y_3 = 1, y_4 = 0, y_5 = 4$. From complementary slackness we have:

$$x_1 \cdot y_4 = 0$$

$$x_2 \cdot y_5 = 0$$

$$x_3 \cdot y_1 = 0$$

$$x_4 \cdot y_2 = 0$$

$$x_5 \cdot y_3 = 0$$

Therefore $x_2 = x_5 = 0$. Replacing in the primal we get

Question 5(d)

$$-2x_1 + x_3 = 4$$

$$-2x_1 + x_4 = -8$$

$$-x_1 = -7$$

Therefore $x_1 = 7$, $x_2 = 0$, $x_3 = 18$, $x_4 = 6$, $x_5 = 0$ and the maximum value of the objective function is -7.